Upper Primary Division

Questions 1 to 10, 3 marks each
1-10 题，每题 3 分

1. A square has area $16\,\text{cm}^2$. What is its perimeter in centimetres?

(A) 4 (B) 8 (C) 12 (D) 16 (E) 24

The square has side length 4 cm therefore perimeter 16 cm,

hence (D).

2. Marlie was born in 2016. In which year will she turn 14?

(A) 2026 (B) 2028 (C) 2030 (D) 2032 (E) 2034

Since $2016 + 14 = 2030$, she will turn 14 in 2030,

hence (C).

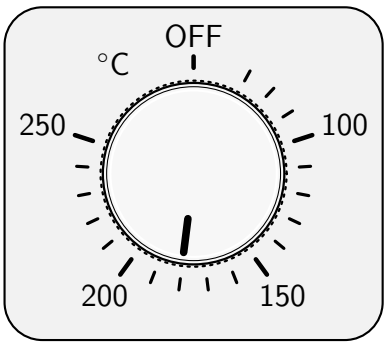
3. At a theme park, one ride costs \$8. How many rides can Sally buy with her \$65?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

The closest multiple of 8 is $8 \times 8 = 64$, so Sally can buy 8 rides,

hence (D).

4. Evan set the temperature dial on his oven as shown. What temperature is it set to?



(A) 155 °C (B) 170 °C (C) 180 °C (D) 220 °C (E) 230 °C

Between 150 and 200 there are 5 intervals, so each interval represents 10 °C. The dial is set 3 intervals above 150 °C which is therefore 180 °C,

hence (C).

5. The following decimals are to be placed in increasing order.

$$0.42 \quad 0.24 \quad 4.02 \quad 2.4 \quad 0.042$$

Which number will be in the middle position?

(A) 0.42 (B) 0.24 (C) 4.02 (D) 2.4 (E) 0.042

When we put the 5 numbers in ascending order, they are $0.042, 0.24, 0.42, 2.4, 4.02$, and so the middle number is 0.42,

hence (A).

6. If I choose one of these cards at random, which of the following am I least likely to get?

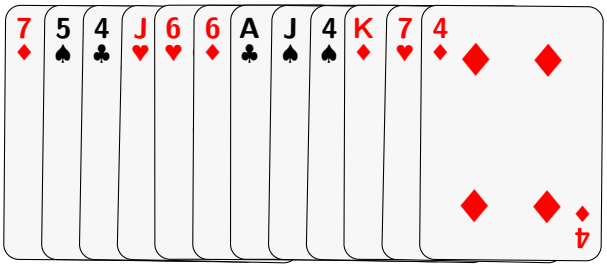
(A) A four

(B) A seven

(C) A spade ♠

(D) A heart ♥

(E) A diamond ♦



For each of spades, diamonds, hearts and fours, there are 3 cards. However, there are only 2 sevens. So I am least likely to get a seven,

hence (B).

7. Leila has found out three facts about the speeds of her four favourite fast birds:

- The spur-winged goose (SWG) is slower than the white-throated needletail (WTN).
- The peregrine falcon (PF) is the fastest bird.
- The frigate bird (FB) is faster than only one of the other birds.

Which of the following lists gives the birds from fastest to slowest?

(A) SWG, WTN, PF, FB
(B) PF, SWG, FB, WTN
(C) PF, WTN, SWG, FB
(D) FB, PF, WTN, SWG
(E) PF, WTN, FB, SWG

Using the abbreviations in the question, place them in order as required. PF is fastest, so must be first. FB is the second slowest, hence third. The other two birds must be 2nd and 4th, but SWG is slower than WTN.

So fastest to slowest is PF, WTN, FB, SWG,

hence (E).

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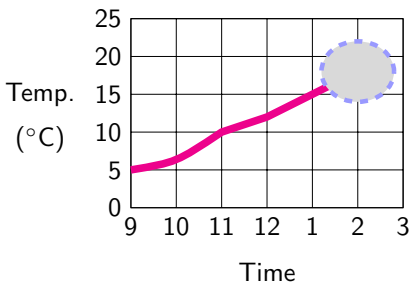
8. This graph shows the temperature throughout the school day, but part of it has been hidden.

The hottest part of the day is at 2 pm. What does the missing section of the graph look like?



(A)
(B)
(C)
(D)
(E)

The temperature rises from 1pm to 2pm but then falls. Picture (D) is the only image showing a rise and then a fall,

hence (D).

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9. In Primeland, they use coins worth \$2, \$3 and \$5.

Which of the following totals is **not** possible using exactly 3 coins?

(A) \$6 (B) \$8 (C) \$10 (D) \$12 (E) \$14

Possible combinations of 3 coins are

$$2+2+2=6 \qquad 2+2+3=7 \qquad 2+2+5=9$$
$$2+3+3=8 \qquad 2+3+5=10 \qquad 2+5+5=12$$
$$3+3+3=9 \qquad 3+3+5=11 \qquad 3+5+5=13 \qquad 5+5+5=15$$

This includes options (A), (B), (C) and (D), but not (E),

hence (E).

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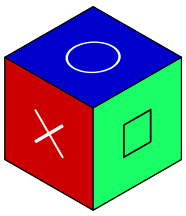
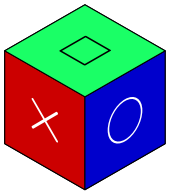
10. The cube pictured has six different faces.

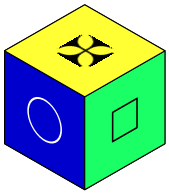
Which of the following is **not** a possible picture of this cube?



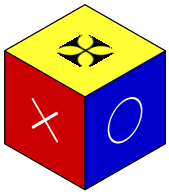
(A)



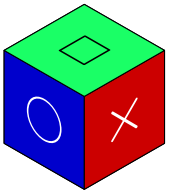
(B)



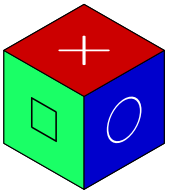
(C)



(D)



(E)

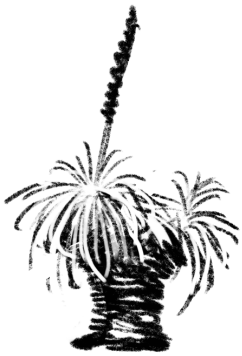


Pictures (D) and (E) are possible by rotating the diagram of the cube by 120° anticlockwise and clockwise, respectively. Picture (A) shows the same faces, but the order of the red-blue-green faces is clockwise in the question and anticlockwise in (A). So (A) is not possible. The other two options are possible: (B) by having the red face downwards and (C) by having the green face downwards. Consequently the only picture that is not possible is (A),

hence (A).

Questions 11 to 20, 4 marks each
11-20 题，每题 4 分

11. The *Balga* is a grass tree found in south-western Australia. Balgas grow between 1 and 2 centimetres taller each year. Approximately how old is a Balga that is 4 metres tall?

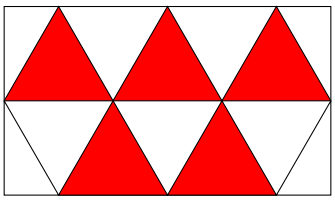


(A) 10 years (B) 30 years (C) 60 years (D) 300 years (E) 1000 years

Each 100 years, a Balga will grow between 1 and 2 metres. To reach 4 metres will take between 4 centuries, at 1 metre per century, or 2 centuries, at 2 metres per century. Of the answers given, 300 years is the best approximation,

hence (D).

12. What fraction of this rectangle is shaded?



(A) $\frac{1}{2}$ (B) $\frac{5}{12}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$ (E) $\frac{5}{9}$

The rectangle has 10 congruent equilateral triangles, and 4 smaller triangles, each being one half the area of the equilateral triangles. So the rectangle is equal to 12 of the equilateral triangles. Since 5 of these are shaded, the fraction shaded is $\frac{5}{12}$,

hence (B).

13. A shop has a special deal where you can buy two different items and get the cheaper item at half price. Roberta bought two of these items, and paid \$16. Which two items did she buy?

(A) A birthday card and a magazine

(B) A gift card and a birthday card

(C) A book and a roll of wrapping paper

(D) A gift card and a roll of wrapping paper

(E) A magazine and a roll of wrapping paper

Since Roberta paid \$16, she spent more than \$8 on the most expensive item, which is either a book or a gift card. If her expensive item was a \$10 book, then she spent \$6 on her other item at half price. But then her cheaper item would have cost \$12, which does not work. So her expensive item can't have been a book.

If her expensive item was a \$12 gift card, she spent \$4 on the other item, which must have been priced at \$8, which is the wrapping paper. So Roberta bought a gift card and a roll of wrapping paper,

hence (D).

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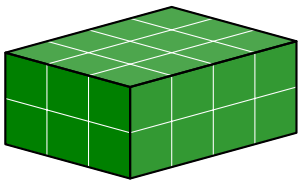
14. A rectangular block measures $4 \times 3 \times 2$ in centimetres. All of its surfaces are painted and then it is cut into 24 cubes of side length 1 centimetre.

How many cubes have exactly one face painted?



(A) 4 (B) 8 (C) 12 (D) 18 (E) 24

In the top layer, only the centre two cubes have a single painted face. Similarly in the bottom layer, the centre two cubes have a single painted face. Consequently, 4 cubes have exactly one painted face,

hence (A).

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15. James convinces his parents to start a new plan for paying him pocket money. In week 1, they will give him 10 cents and every week after that, they will double the amount they pay him.

In which week will he first receive more than \$10 pocket money?

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

Tabulate the amount received each week:

Week	1	2	3	4	5	6	7	8
Amount	\$0.10	\$0.20	\$0.40	\$0.80	\$1.60	\$3.20	\$6.40	\$12.80

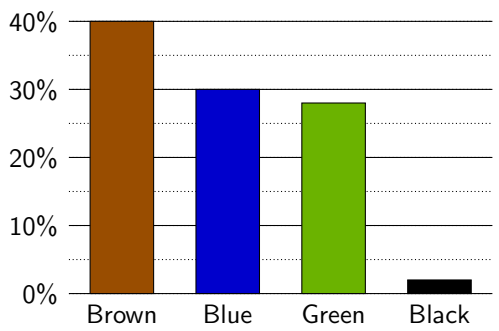
From this table, week 8 is the first week with more than \$10,

hence (C).

16. This graph shows the percentages of different eye colours of the students in Year 6.

In Year 6, 48 students have brown eyes.

What is the total number of students in Year 6?



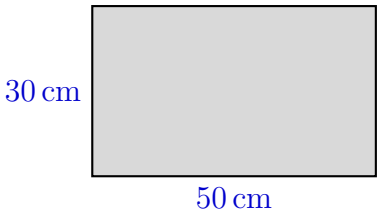
(A) 96 (B) 100 (C) 120 (D) 132 (E) 144

From the graph, 48 students are 40% of the students. So $48 \div 4 = 12$ students is 10% of the students and 120 students is 100% of the students,

hence (C).

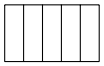
17. Chester has 5 identical tiles, each measuring 10 cm by 30 cm. He wants to tile the area shown with these tiles.

In how many different ways can he do this?

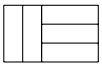


(A) 2 (B) 4 (C) 6 (D) 8 (E) 10

Tiles can be placed horizontally or vertically

One possibility is to have 5 vertical tiles: 

Otherwise the space above or below any horizontal tile must also have horizontal tiles, making a triple of horizontal tiles:

The remaining two tiles must be vertical, and there are 3 ways to arrange these: (i) (ii) (iii) 

So there are 4 different ways to tile the rectangle,

hence (B).

18. I meant to add two numbers together and then subtract a third number.

Unfortunately I got my plus and minus symbols the wrong way round. But I was lucky! The answer still came out to be correct.

Which statement **must** be true?

(A) The first number equals the sum of the other two.

(B) The third number is the difference between the other two.

(C) The first and second numbers are equal.

(D) The second and third numbers are equal.

(E) The third number is zero.

The difference between the two calculations is that the second and third numbers both change from add to subtract, or vice-versa.

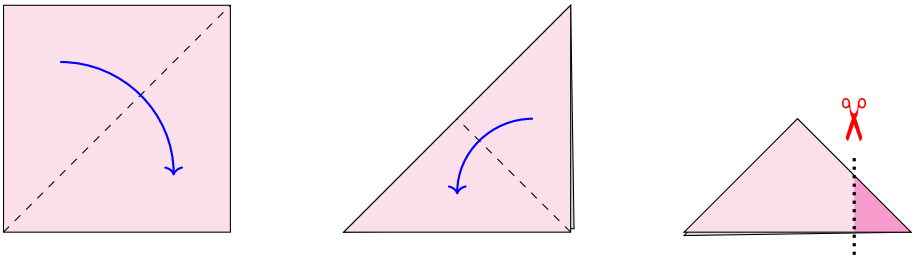
So if the second and third numbers are the same, the answers will both be equal to the first number. If the second number is bigger than the third, then the first calculation would be bigger than the first number and the second calculation would be smaller than the first number, so the two answers will not be the same.

If the second number is smaller than the third, the two answers will not be the same, for similar reasons.

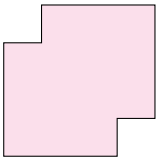
So the only way to have the two answers the same is for the second and third numbers to be the same. So (D) must be true.

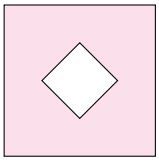
To see that (D) is the only statement that is must be true, a simple example like $5 + 2 - 2 = 5 = 5 - 2 + 2$ is one where (D) is true but (A), (B), (C) and (E) are all false, hence (D).

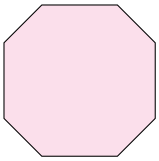
19. A square piece of paper is folded along the dotted lines, as shown in the diagrams. The bottom-right corner is then cut off, as shown by the shading in the third diagram.

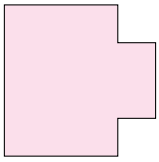


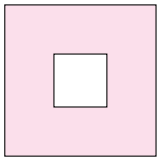
When the paper is unfolded, which of the following is the result?

(A) 

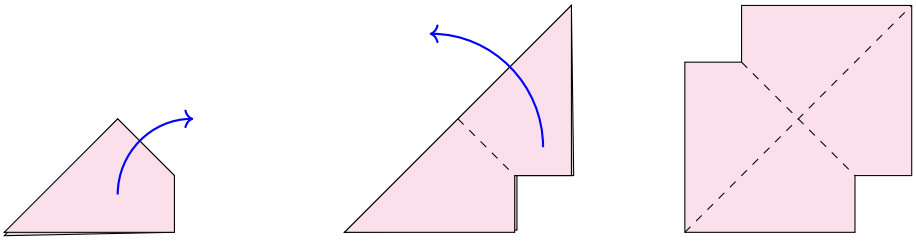
(B) 

(C) 

(D) 

(E) 

Work backwards from the cut shape:



The shape is clearly (A),

hence (A).

20. The number 15 is divisible by its unit digit, that is, $15 \div 5 = 3$.

How many whole numbers between 20 and 50 have this property?

(A) 12 (B) 14 (C) 16 (D) 18 (E) 20

Consider such a number, like 25. Subtract its units digit to make a second number. This number has a units digit equal to 0: $25 - 5 = 20$. Also 20 is a multiple of 5, the units digit.

In general this second number will be a multiple of 10, being one of 20, 30 or 40, and also a multiple of the unit digit.

Going back to the units digit, this will be a factor of this second number. So we make a table of possibilities

Multiple of 10	Possible units digit	Count
20	1, 2, 4, 5	4
30	1, 2, 3, 5, 6	5
40	1, 2, 4, 5, 8	5

This gives 14 numbers: 21, 22, 24, 25, 31, 32, 33, 35, 36, 41, 42, 44, 45 and 48,

hence (B).

Questions 21 to 25, 5 marks each
21-25 题，每题 5 分

21. Troy makes a list of 6 numbers by writing down a starting number then writing the next 5 counting numbers, going up in steps of 1. When he adds them, the total is less than 100.

He increases each number by 1, and the new total is greater than 100.

What is this new total?

(A) 101 (B) 102 (C) 103 (D) 104 (E) 105

Consider the smallest case: $1 + 2 + 3 + 4 + 5 + 6 = 21$. The next case will be $2 + 3 + 4 + 5 + 6 + 7 = 27$ since all 6 numbers have been increased by 1. The next will be $27 + 6 = 33$, then $33 + 6 = 39$ and so on. So when you add six consecutive numbers together, you will always get 21 plus a multiple of 6.

Nearest to 100 are $21 + 78 = 99$ and $21 + 84 = 105$, which can be found to be $14 + 15 + 16 + 17 + 18 + 19 = 99$ and $15 + 16 + 17 + 18 + 19 + 20 = 105$,

hence (E).

22. To prepare a garden bed for growing ginger, Xiao makes a special mix of *ginger soil*. She first mixes 2 buckets of manure with 4 buckets of compost. Then she mixes 1 bucket of this mixture with 3 buckets of sand to make ginger soil.

How much compost is in each bucket of ginger soil?

(A) $\frac{1}{6}$ of a bucket (B) $\frac{1}{8}$ of a bucket (C) $\frac{2}{9}$ of a bucket (D) $\frac{1}{10}$ of a bucket (E) $\frac{1}{12}$ of a bucket

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Alternative 1

After the first mixing, Xiao has 6 buckets. If she then used this to make 6 batches of ginger soil, she would use 18 buckets of sand to make 24 buckets of ginger soil.

These 24 buckets would contain 4 buckets of compost, so that the soil is one-sixth compost. Then each bucket of soil contains $\frac{1}{6}$ of a bucket of compost,

hence (A).

Alternative 2

The mixture of manure and compost is $\frac{1}{3}$ manure and $\frac{2}{3}$ compost. So the final 4 buckets of soil contains $\frac{2}{3}$ of a bucket of compost. One bucket of this soil is $\frac{1}{4}$ of these final 4 buckets, so it contains $\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$ of a bucket of compost,

hence (A).

23. A deck of playing cards originally contained 52 cards. Vicki loses some of these. When she deals the cards she has left equally between four people, she has three cards left over. When she deals them equally between six people, she has one card left over. Which of the following could be the number of cards Vicki lost?

(A) 2 (B) 10 (C) 11 (D) 17 (E) 21

Alternative 1

For each option, consider how many cards would be left. Option A corresponds to 50 cards left. Option B corresponds to 42 cards left. Option C corresponds to 41 cards left. Option D corresponds to 35 cards left. Option E corresponds to 31 cards left. We consider each of these.

When the remaining cards are divided by 4, the remainder is 3. Since $50 \div 4 = 12$ remainder 2, option A is eliminated. Since $42 \div 4 = 10$ remainder 2, option B is eliminated. Since $41 \div 4 = 10$ remainder 1, option C is eliminated. Since $35 \div 4 = 8$ remainder 3, option D is a possible answer. Since $31 \div 4 = 7$ remainder 3, option E is a possible answer.

When the remaining cards are divided by 6, Vicki has one left over. Since $35 \div 6 = 5$ remainder 5, option D is eliminated. This leaves $31 \div 6 = 5$ remainder 1, so option E is the only correct answer,

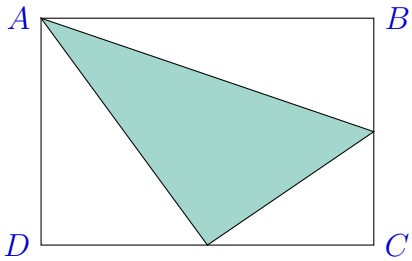
hence (E).

Alternative 2

The number of cards left is 1 more than a multiple of 6 and 3 more than a multiple of 4. Examining multiples of 6, the possibilities that are also 3 more than a multiple of 4 are 7, 19, 31 and 43. This corresponds to losing 45, 33, 21 and 9 cards. Of these, only 21 is given,

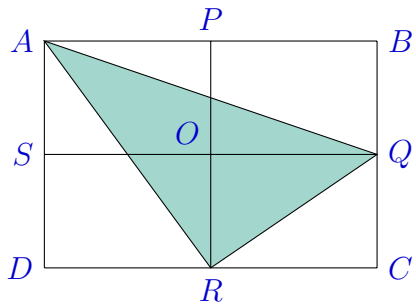
hence (E).

24. The area of this rectangle is 56 square centimetres. The midpoints of BC and DC are joined to A and to each other, to form the shaded triangle shown. What is the area of this triangle in square centimetres?



(A) 14 (B) 21 (C) 24 (D) 25 (E) 28

Join the midpoints of all 4 sides to divide the $ABCD$ into 4 smaller congruent rectangles, each of area $56 \div 4 = 14\,\text{cm}^2$:

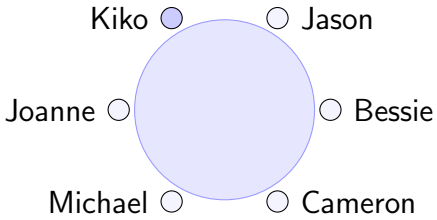


Now consider the areas of the 3 white triangles. Rectangle $APRD$ has area $28\,\text{cm}^2$ so triangle ARD is half this, or $14\,\text{cm}^2$. Similarly, triangle ABQ has area $14\,\text{cm}^2$. Rectangle $OQCR$ has area $14\,\text{cm}^2$, so triangle QCR is half this, or $14 \div 2 = 7\,\text{cm}^2$.

Then triangle AQR has area $56 - 14 - 14 - 7 = 21\,\text{cm}^2$,

hence (B).

25. Kiko and 5 of her friends are seated around a restaurant table.



Each person introduces themself, telling everyone how many of the people at the table they've previously met. Naturally, Kiko has met all 5.

Coincidentally, the other 5 have all met the same number as each other.

Which of the following is **not** a possibility for this number that all 5 of Kiko's friends have met?

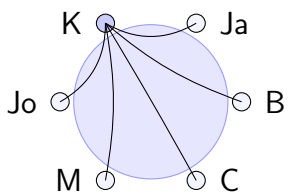
(A) 1 (B) 3 (C) 4 (D) 5

(E) All of (A) to (D) are possible.

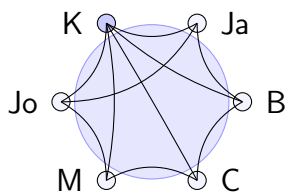
On the diagram, join two people with a line if they know each other. Each line has two ends, so the total number of ends for all lines is even. Kiko has 5 ends, and all 5 of her friends have the same number of ends.

If the 5 friends have an even number of ends each, they have an even number between them, and adding Kiko's 5 ends makes an odd total, which is not possible. On the other hand, if they each have an odd number of ends, they have an odd number, and adding Kiko's 5 makes an even number, which is possible.

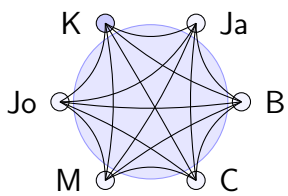
This means that 4 is not a possible number of ends on each of the 5 friends. The odd possibilities 1, 3 and 5 are all possible, as shown below.



$5 + 5 \times 1 = 10$ ends



$5 + 5 \times 3 = 20$ ends



$5 + 5 \times 5 = 30$ ends

So (C) is the only number that is not possible,

hence (C).

For questions 26 to 30, shade the answer as a whole number from 0 to 999 in the space provided on the answer sheet.

问题 26-30 的答案为 0-999 之间的整数，请将答案填涂在答题卡对应区域。

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

第 26-30 题分别为 6, 7, 8, 9, 10 分。

26. Jane has four doting grandparents who each separately tell her that they will help her with her savings plans.

- Granny Ava says, 'Whatever you have when you come to me, I will add \$10.'
- Granny Beth says, 'Whatever you have when you come to me, I will double it.'
- Grampa Charles says, 'Whatever you have when you come to me, I will halve your savings but then add \$50.'
- Grampa Dave says, 'Whatever you have when you come to me, I will double it and add \$5.'

Jane has saved \$10 by herself. She plans to visit each grandparent once, in the order that maximises the amount she has at the end. In dollars, what is this maximum amount?

No matter the order of the visits, the \$10 Jane starts with is doubled twice and halved once, becoming \$20. The other amounts added are \$5, \$10 and \$50. These end up being worth more if they are followed by doubling steps, and no halving steps. In particular, visiting C first means that only the original \$10 is halved, and the \$50 added will later be quadrupled. After visiting C, Jane has \$55.

The second visit is best to be A so that the \$10 is quadrupled later. Jane then has \$65.

The last two visits each double this \$65, but it is better to visit D then B so that the \$5 is doubled once. In this order, Jane has \$135, then \$270,

hence (270).

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27. The numbers 1 to 6 are arranged to form two 3-digit numbers that are then added together. What is the difference between the largest and smallest sums that can be obtained?

The largest sum which can be formed is $642 + 531 = 1173$, by using the largest digits in the larger place values. (Other examples with the same largest sum are possible, such as $632 + 541 = 1173$.) The smallest is $135 + 246 = 381$. The difference between them is $1173 - 381 = 792$,

hence (792).

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28. Skip counting by 5, I write a list of 400 numbers that starts 5, 10, 15 and ends 1990, 1995, 2000.

How many times do I write the digit '0'?

Every second multiple of 5 ends in 0. This gives 200 zeros in the units place.

Among the three- and four-digit numbers, there is a zero in the tens place for the numbers $100, 105, 200, 205, \ldots, 1900, 1905$ and 2000. This adds $19 \times 2 + 1 = 39$ zeros.

Finally, there are extra zeros in the hundreds place for the numbers $1000, 1005, 1010, 1015, \ldots, 1090, 1095$ and 2000. This gives another $2 \times 10 + 1 = 21$ zeros.

Therefore there are $200 + 39 + 21 = 260$ zeros in total,

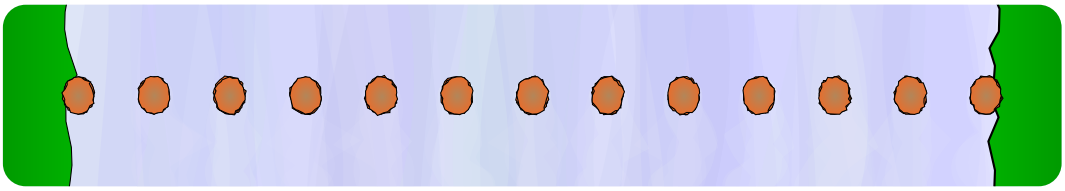
hence (260).

29. Starting with the number 2024, a second number is calculated by squaring each digit and adding these squares together: $2^2 + 0^2 + 2^2 + 4^2 = 24$. The third number is then calculated by adding the squares of the digits of the second number. This continues until the 999th number. What is the 999th number?

The first few numbers are $2024, 24, 20, 4, 16, 37, 58, 89, 145, 42, 20, \ldots$. So the 3rd number is the same as the 11th number. After this the sequence will cycle through these 8 numbers. Since 1000 is a multiple of 8, the 1000th number will be the same as the 8th number. So the 999th number will be the same as the 7th number, which is 58,

hence (58).

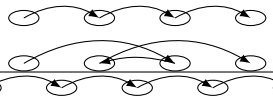
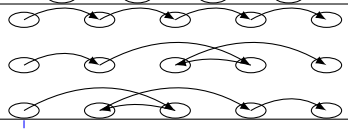
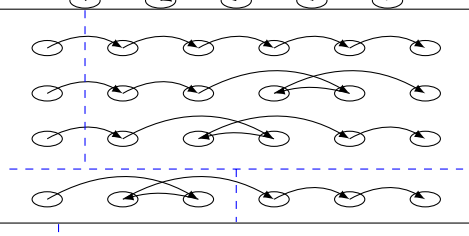
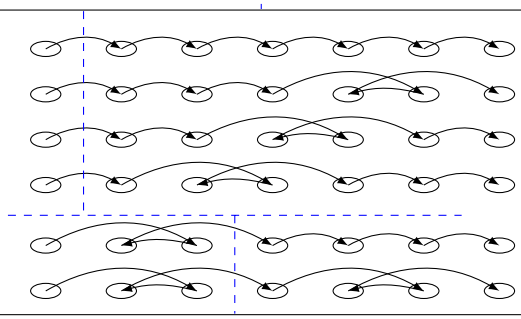
30. To cross a river, there are 13 stepping stones in a line, equally spaced 50 centimetres apart.



I would like to start at the first and finish at the last, and step on each stepping stone exactly once. I can take both 50 centimetre and 1 metre steps, both forwards and backwards.

How many ways are there for me to cross the river?

We work through the cases for the number of stepping stones, until we find a pattern and a rule.

Stones	Possibilities	Count
2		1
3		1
4		2
5		3
6		4
7		6

In this table, the possibilities are grouped by whether they start with a 50 cm step or a 100 cm step. If I start with a 50 cm step, the number of ways to continue is the same as the number of ways to cross with one stone fewer. If I start with a 100 cm step, the next two steps must be 50 cm back then 100 cm forward, and then the number of ways to continue is the same as the number of ways to cross with three stones fewer.

This tells us the pattern in these numbers: $1 + 3 = 4$, $2 + 4 = 6$ and the next one will be

$3 + 6 = 9$. We continue this pattern:

Stones					Count
2					1
3					1
4					2
5	1	+	2	=	3
6	1	+	3	=	4
7	2	+	4	=	6
8	3	+	6	=	9
9	4	+	9	=	13
10	6	+	13	=	19
11	9	+	19	=	28
12	13	+	28	=	41
13	19	+	41	=	60

So there are 60 ways to cross the river like this,

hence (60).